Task 4 — High-Imbalance Clinical Disease Predictor Pipeline

Internship: PROGREE Machine Learning Internship
Project: Breast Cancer Classification using Machine Learning
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Project: Breast Cancer Classification using Machine Learning
Dataset: Wisconsin Breast Cancer Diagnostic Dataset
Tools: Python, Pandas, NumPy, Scikit-Learn, Imbalanced-Learn, XGBoost, Matplotlib and Seaborn

1. Introduction

The objective of this project was to develop a supervised machine learning pipeline for classifying breast cancer diagnostic records as malignant or benign.

In clinical classification, accuracy alone may not fully describe model performance. A model could achieve good overall accuracy while missing cases from an important class. Therefore, recall (sensitivity) was selected as the primary metric for model selection.

The project compares Support Vector Machine (SVM), Random Forest and XGBoost using preprocessing, SMOTE-based class imbalance handling, hyperparameter tuning and five-fold cross-validation.

This project is for educational purposes only and is not intended for clinical diagnosis.

2. Objectives

The main objectives were:

- Load and explore a diagnostic dataset.
- Examine the distribution of benign and malignant cases.
- Separate the data into training and testing sets.
- Apply feature scaling and class imbalance handling.
- Train SVM, Random Forest and XGBoost classifiers.
- Tune model hyperparameters using GridSearchCV.
- Compare models using five-fold cross-validation recall.
- Evaluate the selected model using recall, precision, accuracy, F1-score and ROC-AUC.

- Visualize results using a confusion matrix and comparison graphs.

3. Dataset Description

The Wisconsin Breast Cancer Diagnostic dataset was loaded using Scikit-Learn. It contains diagnostic measurements derived from breast mass images.

The dataset includes 569 records and 30 numerical input features. The target contains two classes: malignant and benign.

For this project, malignant cases were encoded as 1 and benign cases as 0. This made malignant recall the primary positive-class metric.

The dataset contains more benign than malignant records. Although the imbalance is moderate, it provides an opportunity to demonstrate resampling and recall-based evaluation.

4. Data Preprocessing

The data was divided into training and testing sets using an 80:20 split. Stratified splitting was used to preserve the approximate class proportions in both sets.

Feature scaling was applied using StandardScaler. SMOTE was used to create synthetic minority-class examples from the training data.

SMOTE was included inside the machine learning pipeline so that resampling occurred only within each training fold during cross-validation. This approach helps prevent data leakage.

5. Models and Methodology

Three classification algorithms were evaluated.

Support Vector Machine (SVM)

SVM attempts to find a decision boundary that separates the two classes. Feature scaling was included because SVM is sensitive to feature magnitudes.

Random Forest

Random Forest combines multiple decision trees to produce a classification result. It can capture nonlinear relationships between diagnostic features.

XGBoost

XGBoost builds an ensemble of boosted decision trees. Each successive tree contributes to improving the overall model.

Hyperparameter Tuning

GridSearchCV was used to examine multiple parameter combinations for each model. Five-fold cross-validation was performed, and recall was used as the scoring metric.

The model with the highest mean cross-validation recall was selected for final evaluation on the held-out test set.

6. Evaluation Metrics

The following metrics were used:

Accuracy: The proportion of all predictions that were correct.

Precision: The proportion of predicted malignant cases that were actually malignant.

Recall (Sensitivity): The proportion of actual malignant cases correctly identified by the model.

F1-score: The harmonic mean of precision and recall.

ROC-AUC: A measure of how well the model distinguishes between the two classes across classification thresholds.

Recall was the main selection metric because missed malignant cases are an important concern. However, high recall must also be considered alongside false positives and precision.

7. Results and Cross-Validation

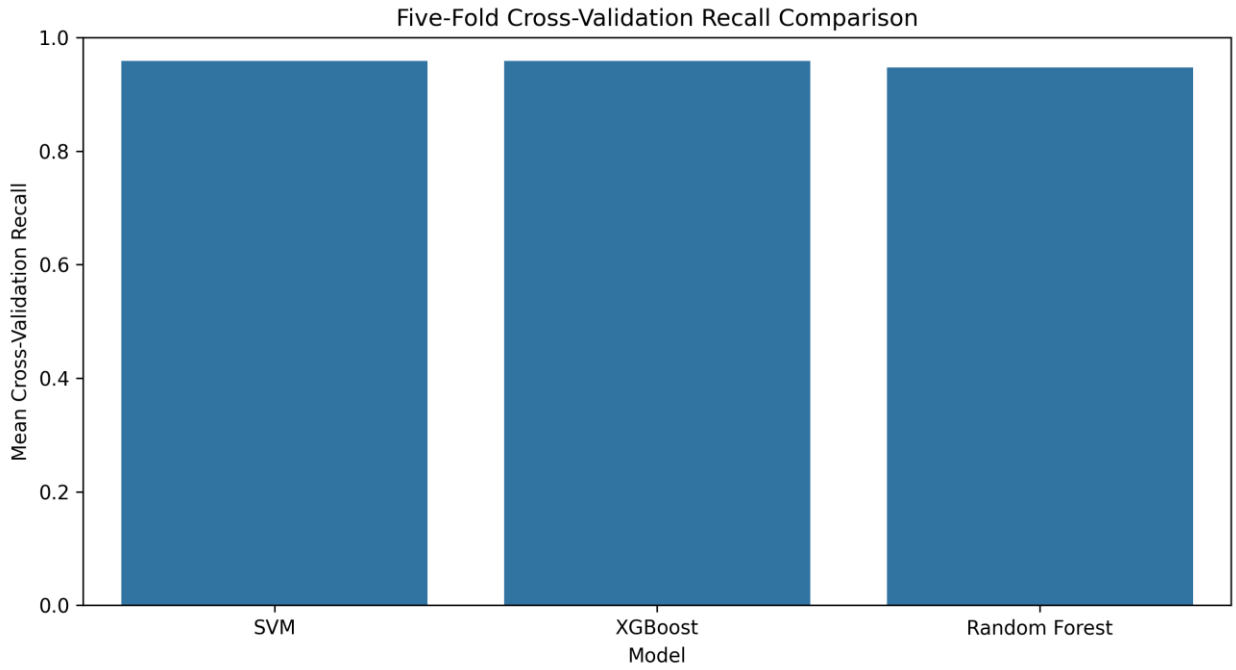
Model Comparison

	Model	Best CV Recall	CV Recall Std	Test Accuracy	Test Precision	Test Recall	Test F1	Test ROC-AUC	Best Parameters
0	SVM	0.9588	0.0353	0.9825	1.0	0.9524	0.9756	0.9934	{'classifier__C': 10, 'classifier__class_weigh...
1	Random Forest	0.9471	0.0432	0.9737	1.0	0.9286	0.9630	0.9960	{'classifier__class_weight': None, 'classifier...

	Model	Best CV Recall	CV Recall Std	Test Accuracy	Test Precision	Test Recall	Test F1	Test ROC-AUC	Best Parameters
2	XGBoost	0.9588	0.0300	0.9649	1.0	0.9048	0.9500	0.9931	{'classifier__learning_rate': 0.1, 'classifier...

The table compares the tuned models using mean cross-validation recall and held-out test metrics.

Five-Fold Cross-Validation Recall



The five-fold results show how recall varied across different training and validation splits. Variation between folds can provide information about the stability of model performance.

Selected Model

Selected model: SVM

Mean cross-validation recall: 0.9588

Test accuracy: 0.9825

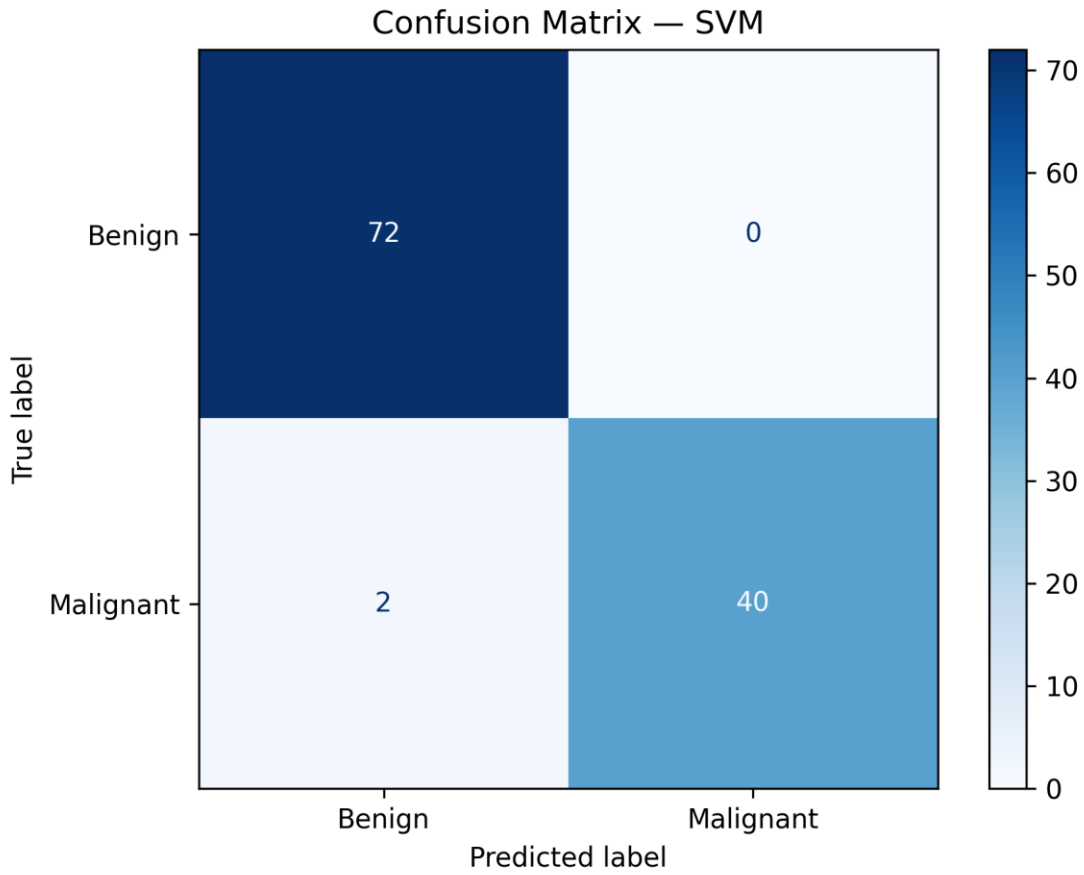
Test precision: 1.0000

Test recall: 0.9524

Test F1-score: 0.9756

Test ROC-AUC: 0.9934

Confusion Matrix



The confusion matrix displays correct and incorrect classifications for benign and malignant cases. In particular, false negatives represent malignant cases classified as benign.

8. Discussion

The model with the highest mean five-fold cross-validation recall was selected for final testing. The test results provide an additional assessment of its performance on data that was not used during model fitting.

The confusion matrix and classification report help explain the types of errors made by the model. A high recall may reduce missed malignant cases but can also lead to more false-positive predictions.

The results should therefore be interpreted using multiple metrics rather than accuracy or recall alone. Performance on this dataset does not establish clinical reliability or generalization to other hospitals and patient populations.

9. Limitations

This project has several limitations:

- The dataset is relatively small and has a moderate rather than extreme class imbalance.
- Synthetic samples created by SMOTE do not replace real clinical observations.
- The dataset may not represent all patient populations or clinical settings.
- Cross-validation results can vary with the data and selected parameters.
- The model has not been validated for medical use.

10. Conclusion

A complete clinical classification pipeline was developed using the Wisconsin Breast Cancer Diagnostic dataset.

The project included exploratory data analysis, feature scaling, SMOTE-based class imbalance handling, training of SVM, Random Forest and XGBoost, hyperparameter tuning, five-fold cross-validation and evaluation on a separate test set.

Recall was used as the main model selection metric, while precision, accuracy, F1-score, ROC-AUC and the confusion matrix provided a broader view of classification performance.

This project provided practical experience in building and evaluating machine learning classification pipelines with a focus on sensitivity and responsible interpretation of diagnostic predictions.

11. Learning Outcomes

Through this project, I learned how to:

- Explore a diagnostic dataset.
- Understand class imbalance.
- Apply feature scaling and SMOTE.
- Build and compare multiple classification models.
- Tune hyperparameters using GridSearchCV.

- Use five-fold cross-validation.
- Evaluate classification models using recall and other metrics.
- Interpret a confusion matrix and classification report.
- Save a trained pipeline and organize a project for GitHub.